

# TFLN\_AI\_NODE\_X2 - Critical Schematic Overview

Dual NCE dies, HBM groups, the TFLN photonic IC, and the RF driver chain are called out here for manufacturer and engineering review. Status: engineering review; complete schematic/netlist source required to close 214 schematic-parity warnings.

|                                                                                                                                                                     |                                                                                                                                                                    |
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| <b>NCE-A DIE</b><br>Ref: U1<br>Value: NCE_Gen3_SpikingBrain<br>Footprint: BGA-2500 40x40mm<br>MPN: LR-NCE-G3-BGA2500-001<br>TFLN lanes: P/N 0..3                    | <b>NCE-B DIE</b><br>Ref: U4<br>Value: NCE_Gen3_SpikingBrain<br>Footprint: BGA-2500 40x40mm<br>MPN: LR-NCE-G3-BGA2500-001<br>TFLN lanes: P/N 0..3                   |
| <b>TFLN PHOTONIC IC</b><br>Ref: U3<br>Value: TFLN_PIC_4xMZM<br>Footprint: Custom optical module 25x8mm<br>MPN: TFLN-MZM-400G-C<br>Function: 4 x TW-MZM / optical TX | <b>RF DRIVER CHAIN</b><br>Refs: U50, U51, U52, U53<br>Value: HMC8410<br>Function: 100GHz differential RF drive<br>Feeds TFLN/MZM modulation chain                  |
| <b>HBM4 FOR NCE-A</b><br>U30, U31, U32, U33, U34, U35<br>6 x HBM4-16GB stacks<br>Group total: 96GB                                                                  | <b>HBM4 FOR NCE-B</b><br>U40, U41, U42, U43, U44, U45<br>6 x HBM4-16GB stacks<br>Group total: 96GB                                                                 |
| <b>OPTICAL OUTPUTS</b><br>U3 outputs: OPT_TX0, OPT_TX1, OPT_TX2, OPT_TX3<br>U1 and U4 expose TFLN_TX lane groups P/N 0..3                                           | <b>ENGINEERING NOTE</b><br>This overview is a readable callout sheet. It does not replace the missing complete schematic/netlist source needed for parity closure. |

| Ref     | Block            | Value / MPN                                   | Role                                                               |
|---------|------------------|-----------------------------------------------|--------------------------------------------------------------------|
| U1      | NCE-A die        | NCE_Gen3_SpikingBrain / LR-NCE-G3-BGA2500-001 | Neural compute engine A; TFLN TX lane group and HBM group U30-U35. |
| U4      | NCE-B die        | NCE_Gen3_SpikingBrain / LR-NCE-G3-BGA2500-001 | Neural compute engine B; TFLN TX lane group and HBM group U40-U45. |
| U3      | TFLN photonic IC | TFLN_PIC_4xMZM / TFLN-MZM-400G-C              | LiNbO3 photonic IC with 4 x TW-MZM and OPT_TX0..OPT_TX3.           |
| U50-U53 | RF driver chain  | HMC8410 / HMC8410LP2FE                        | Differential RF drivers feeding the TFLN/MZM modulation chain.     |
| U30-U35 | HBM4 for NCE-A   | HBM4-16GB / SK_HBM4_16GB_12H                  | Six HBM4 stacks near U1.                                           |
| U40-U45 | HBM4 for NCE-B   | HBM4-16GB / SK_HBM4_16GB_12H                  | Six HBM4 stacks near U4.                                           |